

DesignEval: A Human-Centered Framework for Aligning Graphic Designs with User Intent through Quantitative and Multimodal Evaluation

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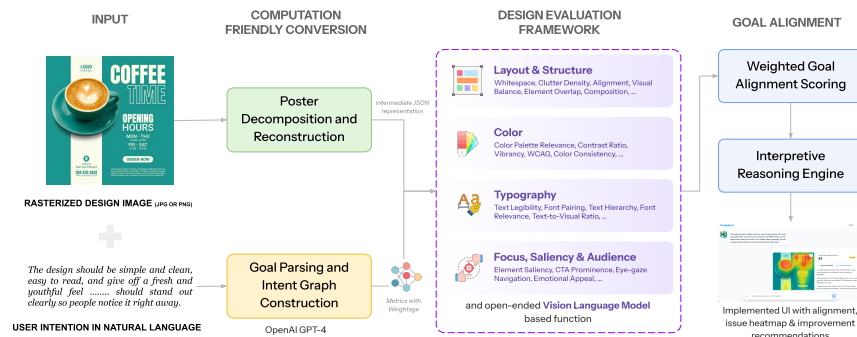
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Graphical Abstract: Aligning Graphic Designs with User Intent at a glance

Abstract

We present DesignEval, a human-centered framework for aligning graphic designs with user intent through quantitative and multimodal evaluation. The system parses user goals into weighted attributes and constraints, which guide evaluation functions for alignment scoring and constraint checking. Interpretability is supported through natural language feedback from vision-language models and heatmaps that visualize the contribution of design elements to goal satisfaction. A user study with 35 designers demonstrates the framework's effectiveness in improving workflow efficiency, creative exploration, and stakeholder communication. DesignEval highlights how AI-driven evaluation can ground design assessment in user goals, enabling iterative refinement and enhancing designer confidence.

Keywords

Graphic Design, Design Evaluation, Human Centered AI, LLMs for Creativity

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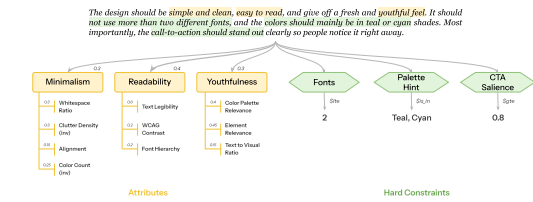
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1 Introduction

Effective design evaluation is needed because aesthetics alone cannot determine whether a design meets its intended goals, and such an evaluation must align visual choices with user intent. Early systems like DesignScape supported novices through interactive optimization [O'Donovan et al. 2015], while generative models such as LayoutGAN learned structured layouts directly from data [Li et al. 2019]. Building on these directions, this research presents DesignEval, a framework for evaluating visual design alignment with user-specified objectives expressed in natural language. The system integrates computer vision, image processing, large language models, and vision-language models to produce both quantitative metrics and qualitative recommendations, establishing a workflow that bridges subjective design intentions with objective computational assessment.

2 Methodology

The DesignEval framework processes rasterized design images and user-defined objectives expressed in natural language. This dual-input approach lets users specify aesthetic, functional, accessibility,



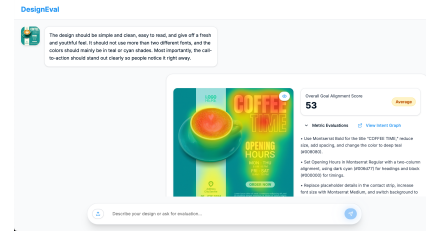
(a) Representing Natural Language Goal as Intent Graph

and audience goals without technical expertise in evaluation metrics. Following input acquisition, the process involves *computation-friendly conversion* through two parallel streams that transform raw inputs into structured representations. The visual decomposition stream [Avudaiappan and Murali 2025] applies computer vision techniques—segmentation and bounding box extraction to isolate elements (such as text, shapes, and images), reconstructing them into a JSON representation with spatial, dimensional, color, typographic, and hierarchical attributes. This combines vision methods with heuristics for color clustering and font estimation. Concurrently, the *goal parsing* stream uses GPT-4 to convert natural language objectives into measurable attributes and constraints, organized into an intent graph (Figure 1a) where nodes represent weighted design attributes. Objectives are split into attributes, mapped to design functions [Murali and Avudaiappan 2024], and hard constraints, extracted as explicit conditions.

The structured representations are assessed through the *design evaluation framework*, a multi-module system jointly processing the reconstructed design and parsed goals. Evaluation covers layout and structure (whitespace, clutter, alignment, balance, overlap), color (palette relevance, contrast, vibrancy, WCAG compliance, consistency), and typography (legibility, hierarchy, font pairing, text-visual ratios). Focus and saliency analysis address call-to-action prominence, element distribution, gaze prediction, and emotional appeal. Vision-language models supplement these with semantic evaluations of subjective qualities such as visual communication of abstract concepts. Evaluation results are processed by the *goal alignment and feedback* system, which aggregates metrics via intent graph weightings to produce alignment scores. Hard constraints are validated independently of the AI.

An interpretive reasoning engine visualizes misaligned regions through heatmaps and generates targeted design recommendations, including adjustments in font, size, and color. The interactive interface translates computational assessments into actionable insights, combining design upload and goal panels with dashboards for alignment scores and detailed metrics (Figure 1b). Attention maps overlay the original design to highlight improvement areas, supported by contextual suggestions for refinement. Interpretability is ensured through explicit links to the intent graph, maintaining transparency behind each recommendation.

This methodology establishes a closed-loop framework for goal-aware design evaluation that integrates computer vision metrics, constraint validation, goal parsing, and vision-language reasoning within a transparent interface. It enables objective scoring and interpretable feedback, uniting computational analysis with practical workflows for systematic, goal-oriented design assessment.



(b) Chat Interface of DesignEval

3 Results & User Feedback

User feedback survey was conducted to evaluate the usefulness of the framework. Conducted with 35 designers having 6 months to 5 years of experience, the responses show consistently strong ratings across all aspects of DesignEval, averaging between 4.28 and 4.5 out of 5. The highest scores were for overall usefulness (4.5) and support in workflow efficiency (4.43), suggesting that DesignEval meaningfully reduces effort in design refinement and decision-making. Users also found it valuable for ideation (4.43) and balancing creativity with constraints (4.29), which highlights its potential in exploring contrasting design directions without losing alignment with objectives. Informal interviews indicated that designers view this tool as a means to validate and justify design decisions, transforming subjective discussions into evidence-based dialogues. By enabling exploration of alternatives and transparent response to feedback, DesignEval enhances the iterative process and strengthens designer-stakeholder collaboration. Ultimately, it embodies the vision of human-centric AI—augmenting creative judgment and fostering confident, efficient design outcomes.

4 Conclusion & Future Direction

This work presented *DesignEval*, a framework that grounds design evaluation in user intent via structured goal parsing, dedicated evaluation functions, and multimodal reasoning. Combining element-level metrics with vision-language interpretation, it outputs alignment scores, issue heatmaps, and targeted suggestions. User feedback points to improved workflow efficiency and clearer communication of design choices. Future work includes multi-user goal parsing, feedback-driven refinement, and standardized benchmarks, advancing AI as a transparent, intent-aware creative collaborator.

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